**Module**: System Programming and Computer Control

**Coursework Title:** LabView Car Wash System(Individual)

**Intake:** APD2F2402CS(AI)

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# Introduction

An overview of the program for vehicle wash controllers

Customers can select and execute several vehicle wash choices using an interactive interface provided by the vehicle Wash Controller Program. It is meant to resemble an automatic car wash's control panel. This software was developed using LabVIEW, a prominent graphical programming environment for instrument control, data collecting, and industrial automation. The project aims to satisfy a number of needs, such as following LabVIEW code standards, offering robust functionality, and creating an easy-to-use user interface.
Objectives

The primary objectives of the Car Wash Controller Program are as follows:

User Interaction: Allow clients to customize their car wash experience by offering them a selection of options.
Automated Control: This technology mimics the sequential processes of an automated car wash based on user selections.
Error Handling: Use robust error handling to deal with scenarios such as the car moving during the wash cycle.
Data Logging: Keep track of the date, time, and options selected during the washing process for later review and analysis.

Crucial Components

Selecting from a Variety of Wash Options: There are several options available to users, such as Air Dry, Spot Free Rinse, Tire Shine Foam, Under Body Wash, Bug Remover, Pre-Soak, High Pressure Wash, and Low Pressure Wax.

In the event that no other options are selected, the program makes sure that the High Pressure Wash option is automatically selected, offering a default wash cycle.

Execution of the Wash Process: The program manages the wash process within a state machine architecture, guaranteeing an understandable and scalable framework.

Based on the user's choices, each wash step is carried out, and real-time location tracking is used to guarantee proper operation of the car.

Error Handling and Position Monitoring: The program continuously monitors the position of the vehicle with a slider control. The automobile will halt the washing operation and alert the user if it moves from its designated place.

The process immediately resumes when the car is parked back in its correct spot.

User Interface: Users can easily select a wash option, start the process, and monitor the status using LEDs and visual indications thanks to a clear and uncomplicated user interface.

The interface responds to user interaction in less than 100 milliseconds, ensuring a smooth and responsive experience.

Data Logging: The application logs the wash details, such as the date, time, and options chosen, in a format that can be read by a spreadsheet at the conclusion of each wash cycle.

Because data is recorded for future reference and analysis, users can review their history of car washes.

Implementation

The Car Wash Controller Program is implemented using the following crucial components:

The controls on the front panel consist of an LED status indicator, Boolean buttons for wash options, and a horizontal pointer slide that simulates the position of the car.

State Machine: Oversees every facet of the washing process, including data logging, user input management, initialization, error handling, and wash step execution.

Event Handling: This involves reacting to user input, like choosing a wash option and initiating the wash.

When no other alternatives are available, Boolean Logic makes sure that the High Pressure Wash option is selected by default.

## Assumptions

Typical Setup for LabVIEW:

Without the use of third-party tools or code produced outside of LabVIEW, the application is developed and tested using a standard LabVIEW system.

Typical Cleaning Option:

To ensure that the default wash cycle is completed in the case that the user does not select any wash options, the High Pressure Wash option is automatically selected.

How to Get a Fixed Wash:

Every wash stage has a five-second time limit, as per the standards.

Position of a Simulated Car:

The position of the car is replicated using a horizontal pointer slide control, and Entry (0), Station 1 (1), Station 2 (2), Station 3 (3), and Exit (4) are indicated with scale markers.

Flexibility of the User Interface:

In order to ensure a fluid and accommodating experience, the user interface reacts to inputs from the user in less than 100 milliseconds.

# System Design (Client/Server)

## Storyboard

Step 1: the costumer will choose the mode he want to put in order to do the washing.

Step 2: the costumer can start the washing and procssed to the next step.

Step 3: the customer will start the washing and clean the car.

Step 4: if the costumer choose under body wash the washing machine will do so.

Step 5: if the costumer choose bug remover the washing machine will do so.

Step 6: if the costumer choose pre-soak the washing machine will do so.

Step 7: if the costumer choose high pressure wash the washing machine will do so.

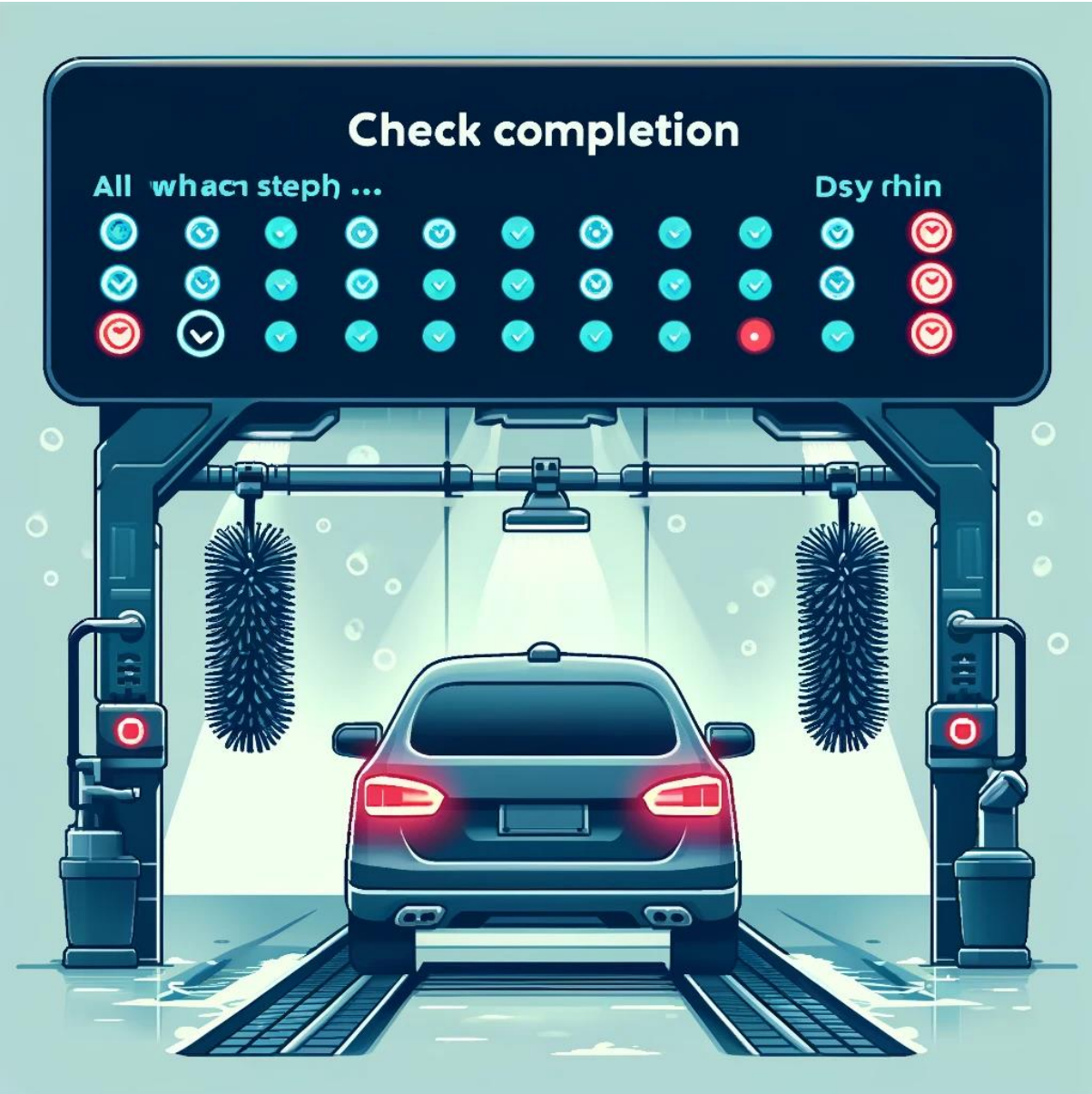
Step 8: if the costumer choose low pressure wax the washing machine will do so.

Step 9: if the costumer choose spot free rinse the washing machine will do so.

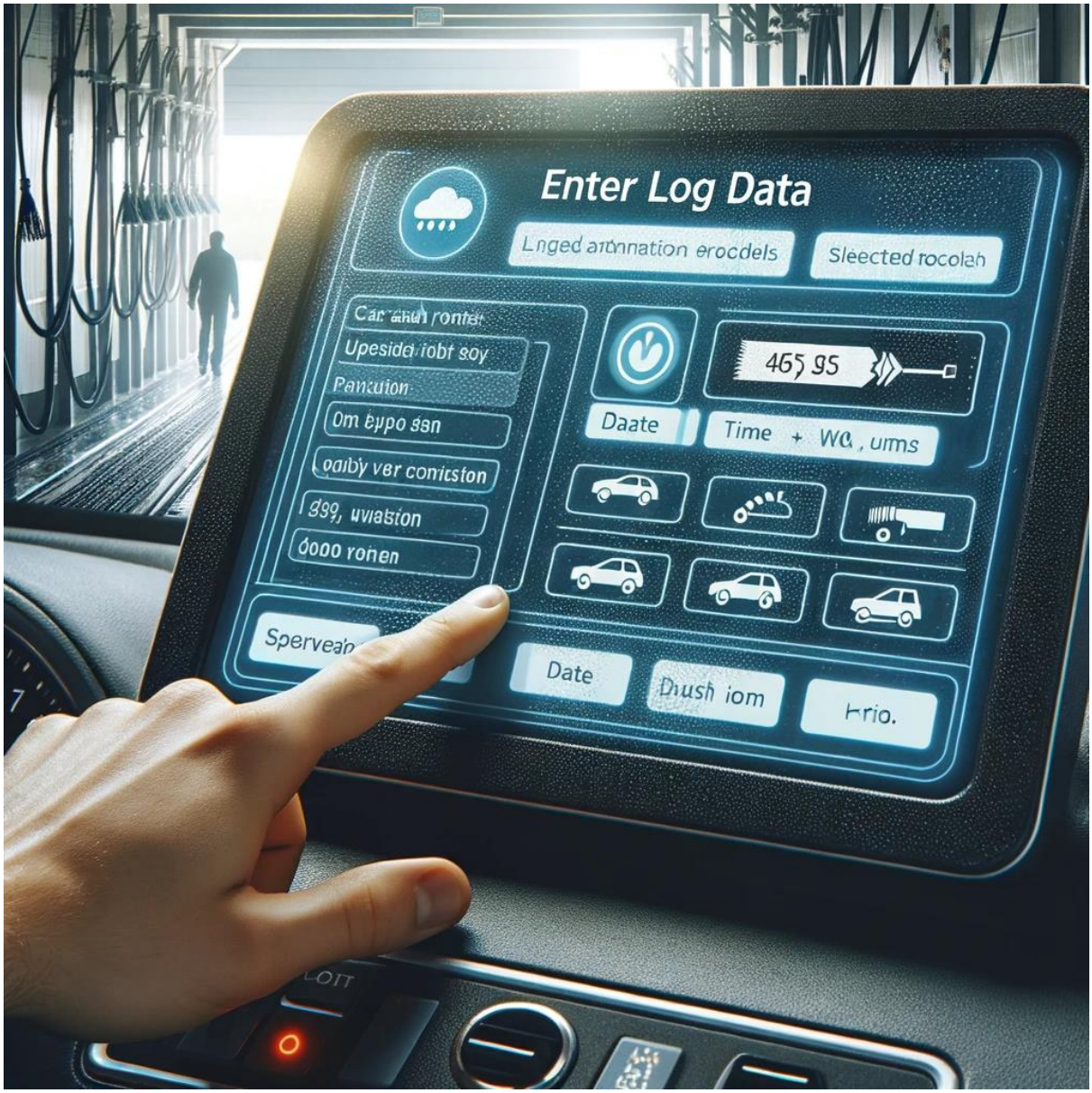
Step 10: if the costumer choose tire shine foam the washing machine will do so.

Step 11: if the costumer choose air dry the washing machine will do so.

Step 12: after all the steps are completed, the car will proceed to go out and check.



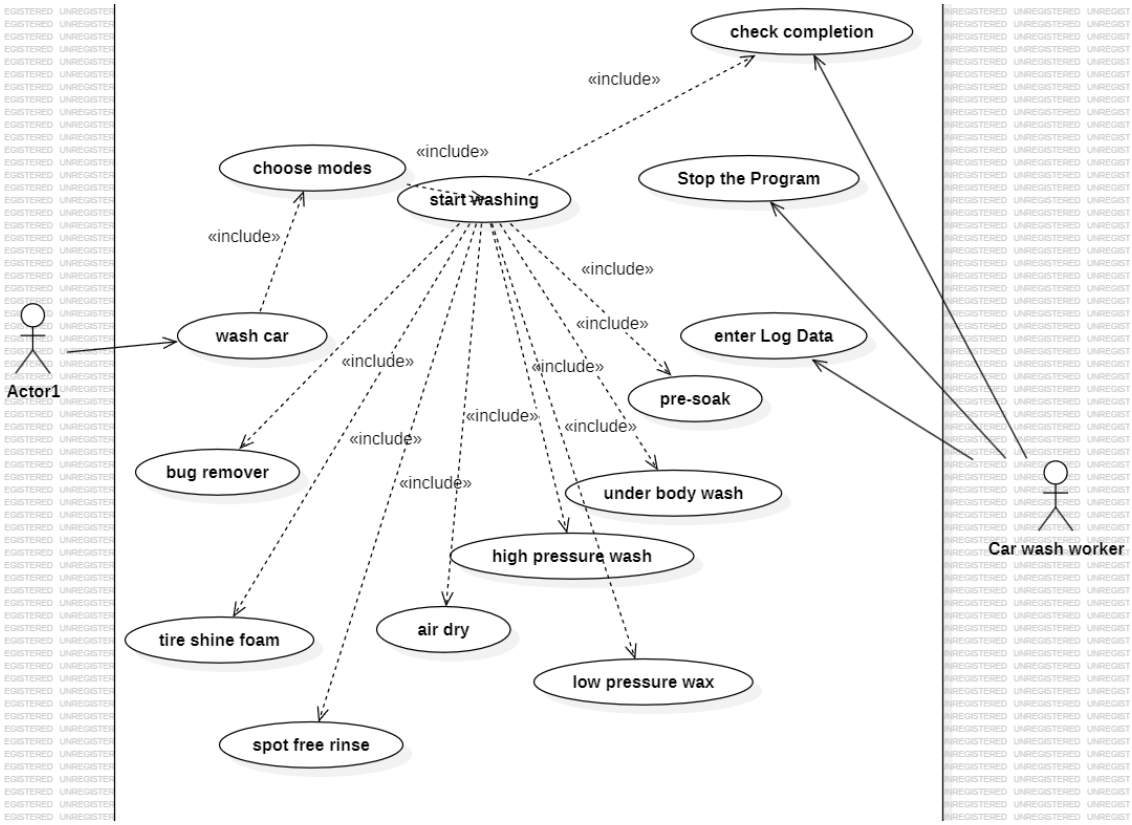

Step 13: the data will be logged into the system.




Step 14: The program will stop completely.

Stop the Program
Trernrayeay 're the mure oiathis
ontay-share toue wash warsh drocess
Ad pme
storsul Sndont
2

## Use-Case Diagrams




## Use case description

| Use Case | Brief Description | Actor | Main Flow |
|---|---|---|---|
| Choose Modes | Allows the user to select different wash modes before starting the wash process. | User | 1. User selects one or more wash options.<br>2. The Car Wash System records the selected options.<br>3. If no options are selected, High Pressure Wash is automatically added. |
| Start Washing | Initiates the car wash process | User | 1. User clicks the Start button. |

| | | | |
|---|---|---|---|
| | based on the selected wash options. | | 2. The Car Wash System verifies the selected wash options.<br>3. The Wash Entry LED changes to indicate the wash is in progress.<br>4. The system transitions to the first selected wash step. |
| • Wash Car | • Performs the overall car wash process, including various wash steps. | • Car Wash System | • 1. The system begins the wash process as per the selected options.<br>2. The system executes each wash step in sequence.<br>3. If the car moves out of position, the process pauses and indicates repositioning.<br>4. The process resumes after the car is correctly positioned. |
| • Under Body Wash | • Cleans the underbody of the vehicle. | • Car Wash System | • 1. The system activates the Under Body Wash if selected. |

| | | | |
|---|---|---|---|
| | | | 2. The underbody spray heads operate for a predefined duration (e.g., 5 seconds).<br>3. The system monitors the car's position to ensure correct operation.<br>4. Upon completion, the system transitions to the next step. |
| • Bug Remover | • Applies a bug remover solution to the vehicle. | • Car Wash System | • 1. The system activates the Bug Remover if selected.<br>2. The bug remover is applied for a predefined duration (e.g., 5 seconds).<br>3. The system monitors the car's position to ensure correct operation.<br>4. Upon completion, the system transitions to the next step. |
| • Pre-Soak | • Applies a pre-soak solution to the vehicle. | • Car Wash System | • 1. The system activates the Pre-Soak if selected.<br>2. The pre- |

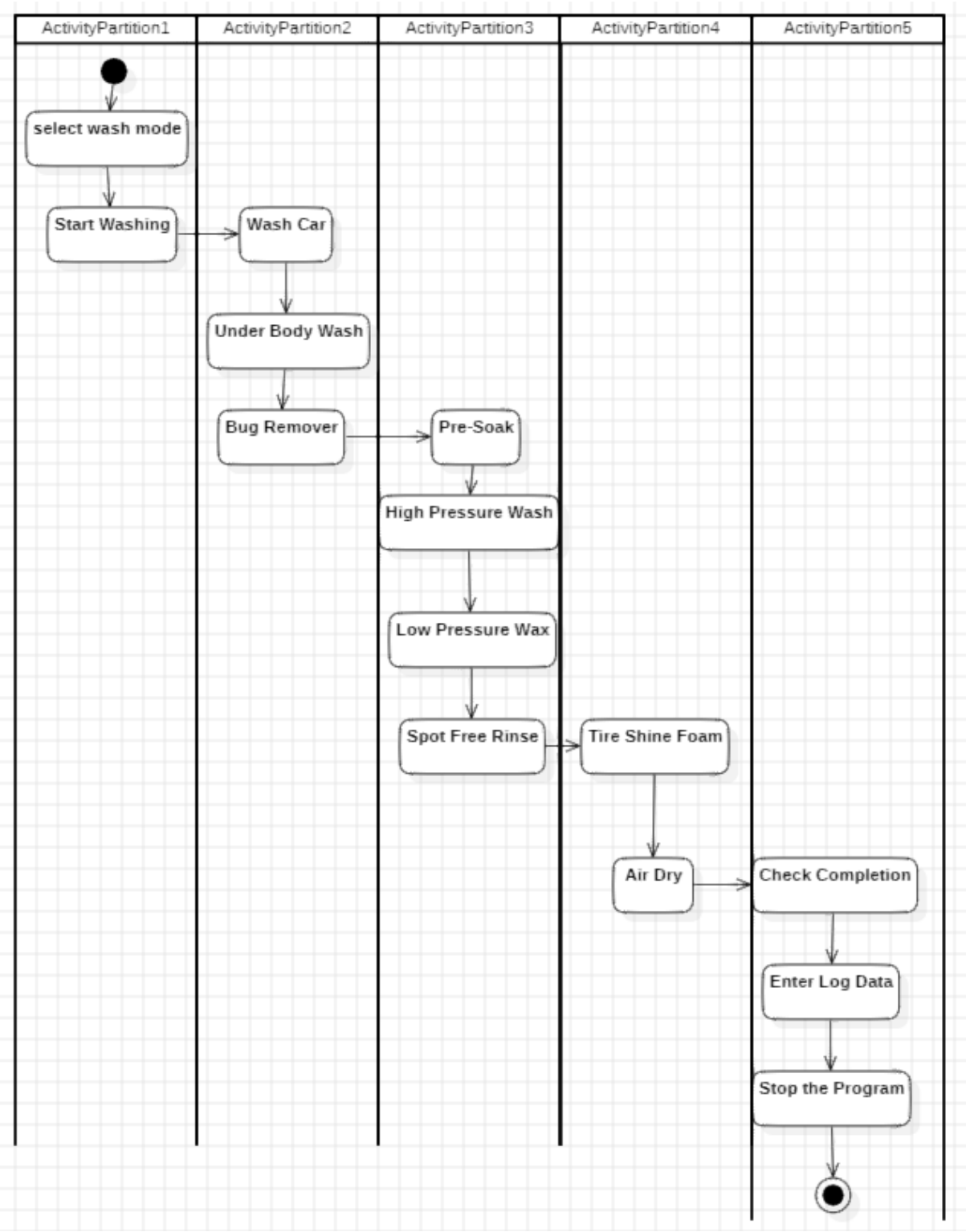
| | | | |
|---|---|---|---|
| | | | soak is applied for a predefined duration (e.g., 5 seconds).<br>3. The system monitors the car's position to ensure correct operation.<br>4. Upon completion, the system transitions to the next step. |
| • High Pressure Wash | • Uses high-pressure water to clean the vehicle. | • Car Wash System | • 1. The system activates the High Pressure Wash if selected.<br>2. The high-pressure wash operates for a predefined duration (e.g., 5 seconds).<br>3. The system monitors the car's position to ensure correct operation.<br>4. Upon completion, the system transitions to the next step. |
| • Low Pressure Wax | • Applies wax to the vehicle using low-pressure spray. | • Car Wash System | • 1. The system activates the Low Pressure Wax if selected. |

| | | | |
|---|---|---|---|
| | | | 2. The wax is applied for a predefined duration (e.g., 5 seconds).<br>3. The system monitors the car's position to ensure correct operation.<br>4. Upon completion, the system transitions to the next step. |
| • Spot Free Rinse | • Rinses the vehicle with spot-free water. | • Car Wash System | • 1. The system activates the Spot Free Rinse if selected.<br>2. The rinse operates for a predefined duration (e.g., 5 seconds).<br>3. The system monitors the car's position to ensure correct operation.<br>4. Upon completion, the system transitions to the next step. |
| • Tire Shine Foam | • Applies shine foam to the tires. | • Car Wash System | • 1. The system activates the Tire Shine Foam if selected.<br>2. The tire shine foam is |

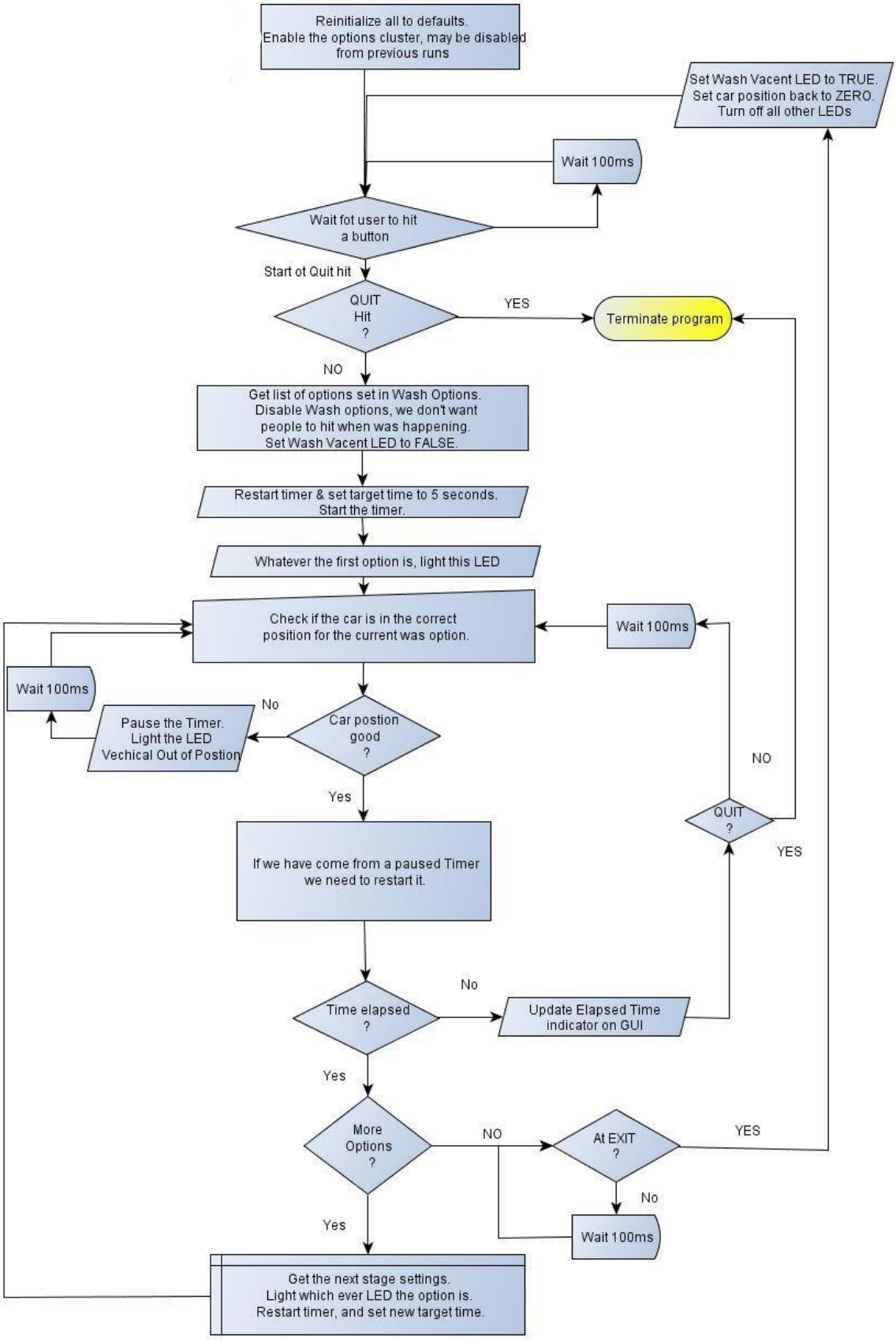
| | | | |
|---|---|---|---|
| | | | applied for a predefined duration (e.g., 5 seconds).<br>3. The system monitors the car's position to ensure correct operation.<br>4. Upon completion, the system transitions to the next step. |
| • Air Dry | • Dries the vehicle using air blowers. | • Car Wash System | • 1. The system activates the Air Dry if selected.<br>2. The air blowers operate for a predefined duration (e.g., 5 seconds).<br>3. The system monitors the car's position to ensure correct operation.<br>4. Upon completion, the system signals the end of the wash process. |
| • Check Completion | • Verifies that all selected wash steps are completed. | • Car Wash System | • 1. The system checks if all selected wash steps have been completed. |

| | | | |
|---|---|---|---|
| | | | 2. If all steps are complete, the system transitions to the log data step or stops the program. |
| • Enter Log Data | • Logs the wash process details. | • Car Wash System | • 1. The system logs the date, time, and selected wash options.<br>2. The log data is stored in a spreadsheet-readable format. |
| • Stop the Program | • Allows the user to stop the car wash operation at any time. | • User | • 1. The user clicks the Stop button.<br>2. The car wash system aborts the current wash operation.<br>3. The system resets all controls and indicators to their default states. |

## Activity Diagrams



## Flowchart


Reinitialize all to defaults.
Enable the options cluster, may be disabled from previous runs
Set Wash Vacent LED to TRUE.
Set car position back to ZERO.
Turn off all other LEDs
Wait 100ms
Wait fot user to hit a button
Start ot Quit hit
QUIT Hit ?
YES
Terminate program
NO
Get list of options set in Wash Options.
Disable Wash options, we don't want people to hit when was happening.
Set Wash Vacent LED to FALSE.
Restart timer & set target time to 5 seconds.
Start the timer.
Whatever the first option is, light this LED
Check if the car is in the correct position for the current was option.
Wait 100ms
Wait 100ms
No
Pause the Timer.
Light the LED
Vechical Out of Postion
Car postion good ?
Yes
NO
QUIT ?
YES
If we have come from a paused Timer we need to restart it.
No
Time elapsed ?
Update Elapsed Time indicator on GUI
Yes
More Options ?
NO
At EXIT ?
YES
No
Yes
Wait 100ms
Get the next stage settings.
Light which ever LED the option is.
Restart timer, and set new target time.

## Protocol Design (Client/Server)

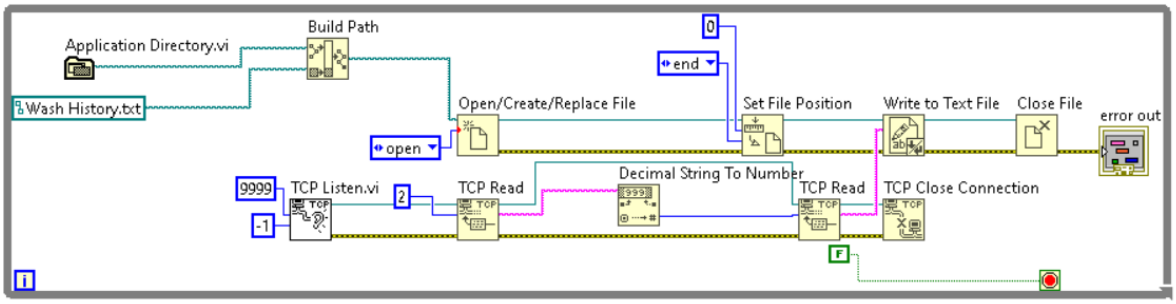



**Figure. Server**

The program is set up in this particular case to listen on port 9999 for data (Date\Time\Wash Type) that the client sends. "TCP Listen.vi," the left VI, keeps an eye on incoming connections on a TCP socket. After connecting, the right VI, called "TCP Read.vi," reads two bytes of data from the client and appends it to a file called "Wash History.txt" along with the current data (Date\Time\Wash Type).

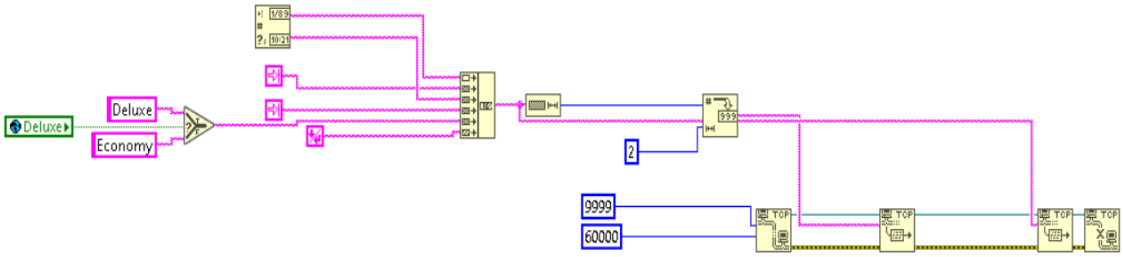



**Figure. Client**

By documenting the details of the wash type the user has selected, as well as the time and date of the selection, the client fulfills an essential role. The client utilizes port 9999 to actively deliver this data to the server when it has obtained it. This organized approach ensures that the server receives relevant and timely information about user preferences and allows for the centralized storing and analysis of prior wash data. Using the client-server approach, which allows the client to initiate the data collection process and forward the collected information to the server, makes managing user-specified wash settings simple and efficient.

## Health and Safety Considerations

Fall and Slip Risks:

To avoid slipping, use flooring materials with good traction.
Post obvious cautionary notices regarding moist floors.
Check and clean frequently to avoid soap or water accumulation.

Exposure to chemicals:

Make use of vetted and safe cleaning agents.

Ascertain adequate ventilation to release odors.
Observe PPE requirements, which include providing workers with gloves and eye protection.

Positioning of the Vehicle:

For exact alignment, mark and ignite the vehicle positioning switches.
Include sensors to track a car's location and halt wash cycles as needed.
For added safety, install an emergency stop device.

Procedures for Emergencies:

Establish and publish emergency protocols.
Install emergency stop buttons that are conveniently accessible.
Provide operators with frequent drills and instruction.

Security of File Logging:

Put in place reliable procedures for data storage and transmission.
Audit and check the logging system frequently for security flaws.
Assure the security of client data against unwanted access.

Failures due to mechanics:

Put in place a schedule for preventive maintenance.
Plan ahead and replace worn parts as needed.
Regular inspections can help to spot such problems early and take appropriate action.

Problems Affected by the Weather:

Install hardware that is resistant to the weather.
Build the system with the surroundings in mind.
Make sure you have contingency plans in place for severe weather, such as storm-related temporary shutdowns.

## Implementation

Central Processing Unit: A PLC (Programmable Logic Controller), also known as a programmable microcontroller, that has the memory and processing capacity to handle control tasks in real time.
Sensor Network: Using cameras or proximity sensors to identify the dimensions and entry of vehicles.
Water flow sensors are integrated to track and modify the strength of the water spray.
Water Distribution System:
Make use of water pumps with enough capacity and high pressure.
Employ nozzles with movable settings to precisely regulate overspray patterns.
Communication Equipment: To ensure a smooth LabVIEW integration, include Ethernet or other communication interfaces.
wireless modules are incorporated to provide remote control and monitoring.

# Conclusion

Restrictions/Points to Take

Fall and Slip Hazards: There is always a chance of falling and slipping, even with safety precautions.
It's crucial to monitor frequently and act quickly when spills occur.

Chemical Exposure: Even with protective gear and ventilation, extended chemical exposure can be harmful to one's health.
Getting regular checkups and reducing the use of toxic chemicals can be beneficial.

Vehicle Positioning: Emergency stop systems and sensors are not infallible.
Testing and routine maintenance are essential.

Procedures for Emergencies: Employees might not always react appropriately in times of crisis.
Response can be enhanced with regular drills and new instruction.

Security of File Logging: Cybersecurity risks are ever-changing.
Security protocols must be updated on a regular basis.

Mechanical Failures: Time-consuming and expensive preventive maintenance is sometimes necessary.
It's crucial to strike a balance between maintenance and downtime expenses.

Weather-Related Challenges: Severe weather still has the potential to cause operational disruptions.
Thorough preparations for inclement weather can reduce disruptions.

upcoming improvements

Automated Diagnostics: Put in place systems for automatically identifying and documenting problems.

Improved User Interface: Provide a touch-screen interface with real-time feedback and controls that is easier to use.

Mobile Integration: Provide a mobile app for customers to choose and pay for their preferred wash options.

Green initiatives include using less water and introducing eco-friendly wash alternatives with biodegradable soaps.

Predictive Maintenance: Make proactive maintenance plans by utilizing AI to forecast component failures.

Enhanced Safety Features: To improve safety, including automated alerts and real-time video surveillance.

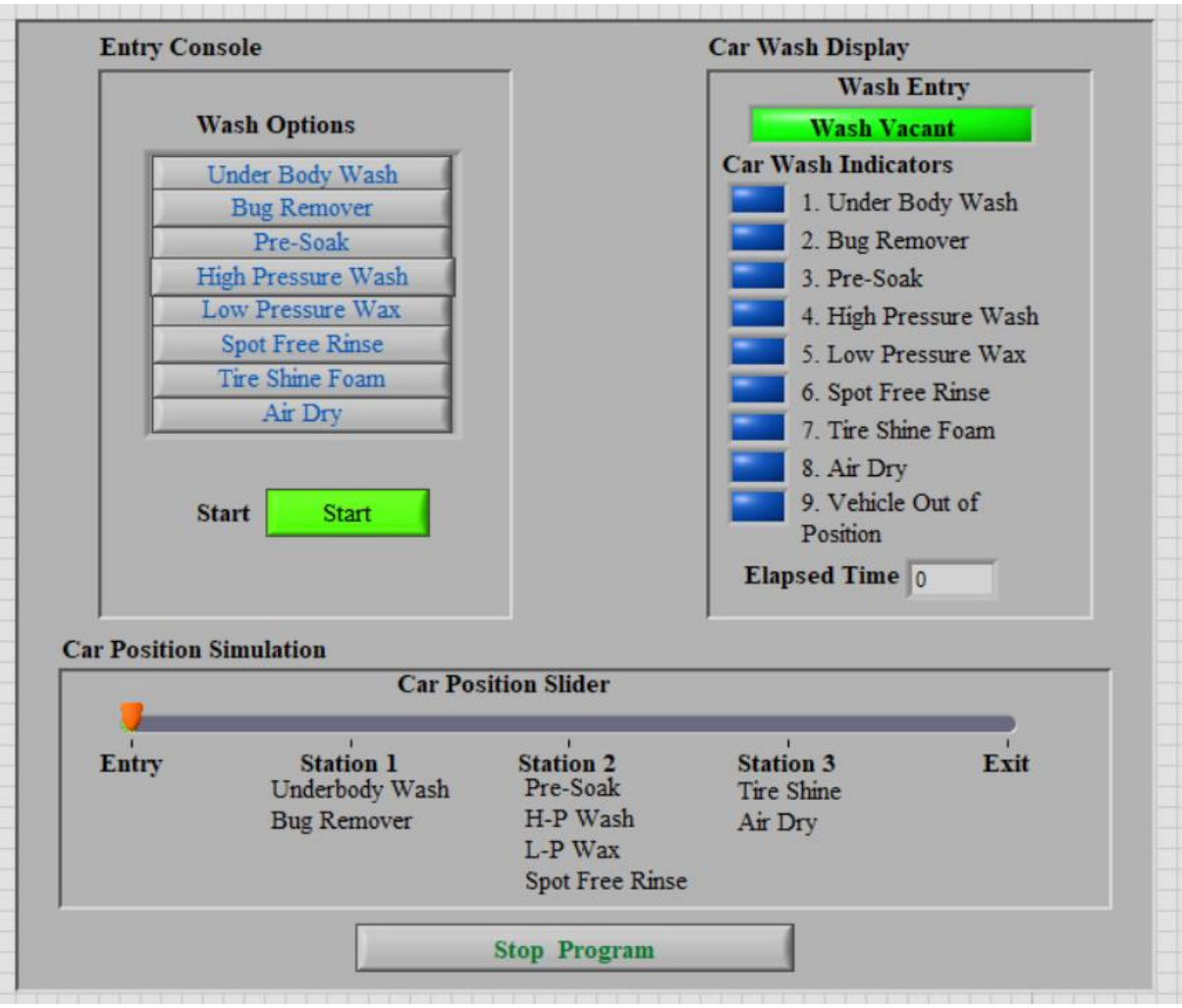
Energy Efficiency: Make investments in renewable energy sources and energy-efficient machinery.

Increased Services: Provide extra services like tire pressure checks and interior cleaning.

## References

Product. (2023). Ni.com. https://www.ni.com/en-my/shop/product/labview.html

## Appendix



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**Getting Started**

1. **Launch the Application**:
   - Open the LabVIEW environment.
   - Load the Car Wash Controller Program VI.
   - Run the VI to start the application.
2. **Initial System State**:
   - The system initializes with all controls enabled.
   - The Wash Entry LED is green, indicating the car wash is vacant.
   - All Car Wash Indicators are off, and the Elapsed Time displays 0.00.
   - The Car Position Slider is set to the Entry position.

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**User Interface Overview**

1. **Wash Options**:
   - **Buttons**: Under Body Wash, Bug Remover, Pre-Soak, High Pressure Wash, Low Pressure Wax, Spot Free Rinse, Tire Shine Foam, Air Dry.
   - **Function**: Select the desired wash options. These will remain selected until the completion of the wash cycle.
2. **Start Button**:
   - **Function**: Initiates the car wash process based on the selected wash options.
3. **Car Position Slider**:
   - **Function**: Simulates the car's position during the wash process. Positions include Entry (0), Station 1 (1), Station 2 (2), Station 3 (3), Exit (4).
4. **Stop Button**:
   - **Function**: Stops the wash process at any time and resets the system.

5. **Indicators**:
    - **Wash Entry LED**: Green for vacant, red for in progress.
    - **Car Wash Indicators**: LEDs for each wash step.
    - **Elapsed Time**: Displays the time elapsed for each wash step.
    - **Vehicle Out of Position LED**: Indicates if the vehicle is not in the correct position.

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**Operating Instructions**

1. **Choose Modes**:
    - Select one or more wash options using the Wash Options buttons.
    - If no options are selected, the system will automatically include High Pressure Wash.
2. **Start Washing**:
    - Click the Start button.
    - The system will verify the selected options.
    - The Wash Entry LED changes to red, indicating the wash is in progress.
    - The system transitions to the first selected wash step.
3. **Wash Car**:
    - The car goes through the wash process as per the selected options.
    - Each wash step executes sequentially.
    - The system monitors the car's position using the Car Position Slider.
    - If the car moves out of position, the process pauses, and the Vehicle Out of Position LED turns on.
    - The process resumes once the car is repositioned correctly.

4. **Check Completion**:
   - The system verifies that all selected wash steps are complete.
   - If all steps are complete, the system transitions to log the data.
5. **Enter Log Data**:
   - The system logs the date, time, and selected wash options into a file.
   - The log file is stored in a spreadsheet-readable format.
6. **Stop the Program**:
   - Click the Stop button to abort the wash process at any time.
   - The system resets all controls and indicators to their default states.

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**Maintenance and Troubleshooting**

1. **Regular Maintenance**:
   - Ensure all sensors and indicators are functioning correctly.
   - Regularly check and replace worn-out parts.
2. **Troubleshooting**:
   - **Car Position Errors**: Ensure the Car Position Slider is functioning and the car is correctly positioned.
   - **System Failures**: Restart the program and check for any loose connections or faults in the system.
3. **Emergency Procedures**:
   - In case of an emergency, use the Stop button to halt the process immediately.

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**Safety Considerations**

1. **Slip and Fall Hazards**:
   - Ensure the floor is dry and warning signs are in place.
   - Regularly inspect and clean the area to prevent slip hazards.
2. **Chemical Exposure**:
   - Use approved cleaning chemicals and proper ventilation.
   - Ensure workers wear PPE, including gloves and eye protection.
3. **Mechanical Safety**:
   - Regularly inspect mechanical components and sensors.
   - Use emergency stop mechanisms in case of mechanical failures.

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